

5.4 Discrete random variables

Probability and Statistics 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A probability distribution has nonnegative probabilities summing to 1. Expected value is $\sum xP(X=x)$; variance is $\sum x^2P(X=x) - [E(X)]^2$. A binomial model requires fixed number of independent trials with constant success probability and two outcomes per trial.

Worked example: follow the reasoning

For $X \sim \text{Bin}(4, 0.5)$, $E(X) = np = 2$ and $\text{Var}(X) = np(1-p) = 1$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 For $P(X=0)=0.2$, $P(X=1)=0.5$, $P(X=2)=0.3$, find $E(X)$ and $\text{Var}(X)$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$E=0+0.5+0.6=1.1$ [1]; $E(X^2)=0.5+1.2=1.7$ [1]; $\text{Var}=1.7-1.1^2=0.49$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.